

# Observation Record - PBL Tasks

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**Week:** 6 (Advanced Node & Express)

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## Q1. Functions vs Classes in JavaScript

### 1. Definition / Introduction

This task explores functions vs classes in javascript and demonstrates how they are practically applied in full stack development.

### 2. Key Points

- Implementation of core concepts.
- Testing and debugging.

### 3. Working / Explanation

The code executes sequentially to fulfill the task requirements, logging the outputs clearly.

### 4. Examples / Code

```
// Source Code attached in src/task1.js
```

### Output Screenshot

```
C:\Users\anits-csm>cd C:\Users\anits-csm\Desktop\NodeModule  
  
C:\Users\anits-csm\Desktop\NodeModule>node app.js  
30  
  
C:\Users\anits-csm\Desktop\NodeModule>|
```

## Test Case 1

```
C:\Users\anits-csm>cd C:\Users\anits-csm\Desktop\NodeModule  
C:\Users\anits-csm\Desktop\NodeModule>node app.js  
30  
C:\Users\anits-csm\Desktop\NodeModule>
```

## Test Case 2

```
C:\Users\anits-csm>cd C:\Users\anits-csm\Desktop\NodeModule  
C:\Users\anits-csm\Desktop\NodeModule>node app.js  
30  
C:\Users\anits-csm\Desktop\NodeModule>
```

## 5. Conclusion

Successfully implemented and tested functions vs classes in javascript.

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## **Q2. Node.js Built-in Modules**

### 1. Definition / Introduction

This task explores node.js built-in modules and demonstrates how they are practically applied in full stack development.

### 2. Key Points

- Implementation of core concepts.
- Testing and debugging.

### 3. Working / Explanation

The code executes sequentially to fulfill the task requirements, logging the outputs clearly.

## 4. Examples / Code

```
// Source Code attached in src/task2.js
```

### Output Screenshot

```
Node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js
----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4
|
```

### Test Case 1

```
Node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js

----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4
```

## Test Case 2

```
Node.js v26.8.1

C:\Users\anits-csm\Desktop\NodeModule>node modules.js
----- OS MODULE -----
Platform: win32
Architecture: x64
Total Memory: 8262455296
Free Memory: 648368128
Home Directory: C:\Users\anits-csm

----- PATH MODULE -----
Directory: C:\Users\anits-csm\Desktop\67
File Name: me.text.txt
Extension: .txt
Joined Path: C:\Users\anits-csm\Desktop\NodeModule\data\file.txt

----- DNS MODULE -----

----- NET MODULE -----
Server running on port 3000
IP Address: 192.178.193.138
IP Family: 4
```

## **5. Conclusion**

Successfully implemented and tested node.js built-in modules.

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## **Q3. NPM and Third-Party Packages**

### **1. Definition / Introduction**

This task explores npm and third-party packages and demonstrates how they are practically applied in full stack development.

### **2. Key Points**

- Implementation of core concepts.
- Testing and debugging.

### **3. Working / Explanation**

The code executes sequentially to fulfill the task requirements, logging the outputs clearly.

### **4. Examples / Code**

```
// Source Code attached in src/task3.js
```

## Output Screenshot



## Test Case 1



## Test Case 2



## **5. Conclusion**

Successfully implemented and tested npm and third-party packages.

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## **Q4. Express.js Middleware**

### **1. Definition / Introduction**

This task explores express.js middleware and demonstrates how they are practically applied in full stack development.

### **2. Key Points**

- Implementation of core concepts.
- Testing and debugging.

### **3. Working / Explanation**

The code executes sequentially to fulfill the task requirements, logging the outputs clearly.

### **4. Examples / Code**

```
// Source Code attached in src/task4.js
```

## **Output Screenshot**

```
C:\Users\anits-csm\Desktop\NodeModule>node server.js
Server is running at http://localhost:3000
Request received from client
Request received from client
|
```

### Test Case 1

```
C:\Users\anits-csm\Desktop\NodeModule>node server.js
Server is running at http://localhost:3000
Request received from client
Request received from client
|
```

### Test Case 2

```
C:\Users\anits-csm\Desktop\NodeModule>node server.js
Server is running at http://localhost:3000
Request received from client
Request received from client
```

## **5. Conclusion**

Successfully implemented and tested express.js middleware.

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